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Preventing or Eradicating Factor VIII Antibody Formation in Patients with Hemophilia A: What Can We Learn from Other Disorders?

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Abstract

Eradiation of factor VIII (FVIII) specific neutralizing antibodies (also known as inhibitors) by the traditional method of immune tolerance induction (ITI) is costly and unsuccessful in one out of three patients. Furthermore, effective inhibitor prevention strategies are presently lacking. An overview is given in this narrative review of antidrug antibody prevention or eradication strategies that have been used in disorders beyond hemophilia A, with the aim of analyzing what we can learn from these strategies for hemophilia A. Prevention of antidrug antibody formation using rituximab, methotrexate, and intravenous immunoglobulins in patients with Pompe's disease seems effective but carries a high risk of adverse events. Based on studies in patients with rheumatoid arthritis and inflammatory bowel disease, it seems likely that treatment with methotrexate alone would also be able to prevent inhibitor formation in hemophilia A patients. Besides side effects, it is unclear whether immune tolerance to FVIII would persist after cessation of immunomodulatory therapy with methotrexate. A combination of cyclophosphamide and corticosteroids, used to treat antibody-mediated pure red cell aplasia, could be further investigated to eradicate inhibitors in hemophilia A patients who are refractory to ITI. In summary, insights gained from research on antidrug antibody formation in other diseases could be helpful in devising alternative treatment strategies for inhibitor development.

Keywords

- inhibitor formation
- immunomodulation
- antidrug antibodies

Hemophilia A is a hereditary X-linked hemorrhagic disorder that is caused by genetic mutations in the *F8* gene. These mutations lead to a deficiency of functional clotting factor VIII (FVIII), which is associated with frequent bleeds, especially in joints and muscles. In the long term, repeated joint bleeds cause bleeding-induced joint damage with concomi-

tant disability and reduced quality of life. The disease can be treated by intravenous administration of the deficient factor with FVIII concentrates. The severity of the disease is based on the plasma concentration of clotting factor and is usually classified as severe (< 0.01 IU/mL), moderate (0.01–0.05 IU/mL), or mild (> 0.05–0.40 IU/mL). Compared to patients with

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mild and moderate hemophilia A, patients with severe hemophilia A experience more frequent bleeding episodes. In addition, most bleeds in patients with mild or moderate hemophilia A are due to trauma or surgery, whereas the majority of bleeds in those with severe hemophilia A occur spontaneously (i.e., are nontraumatic bleeds).¹

Inhibitor Formation

A major treatment complication in hemophilia A is the formation of neutralizing antibodies against FVIII, also known as inhibitors (because they inhibit the function of FVIII), which renders subsequent treatment with FVIII ineffective. Inhibitor formation is most common in patients with severe hemophilia A, as roughly one in three of these patients develop clinically relevant inhibitors.² There is a strong relationship between the incidence of inhibitor formation and the number of days that a patient is exposed to treatment with FVIII (also referred to as the number of exposure days). In patients with severe hemophilia A, inhibitors develop after a median of 15 exposure days,³ and almost all inhibitors occur within the first 75 exposure days.^{4,5} Due to the relatively severe bleeding phenotype of these patients, they are exposed to FVIII at a very young age, especially if prophylactic treatment with FVIII is initiated. Consequently, most severe hemophilia A patients develop inhibitors at a very young age. The median age at which inhibitors were detected was 1.3 years (interquartile range [IQR]: 1–2) in a European registry of 108 severe hemophilia A patients.⁶ In patients with neutralizing antibodies, normal doses of FVIII concentrates are no longer effective as prevention or treatment for bleeding. Therefore, these patients need to be treated with FVIII bypassing agents such as recombinant activated FVII (rFVIIa) or activated prothrombin complex concentrate (aPCC).⁷

Immune Tolerance Induction

To eradicate inhibitors in hemophilia A patients, frequent administration of high doses of FVIII over a long period of time is needed. This is commonly known as “immune tolerance induction” (ITI).⁸ Well-known ITI protocols include the “Bonn” protocol (basic protocol: 100–150 IU/kg FVIII twice daily)⁹ and the “van Creveld” protocol (starting dose is 25 IU/kg FVIII every other day; dosage is decreased when FVIII recovery exceeds 30%).¹⁰ The required duration of ITI to obtain tolerance to FVIII varies per patient. In a large multicenter randomized clinical trial comparing a high (200 IU/kg/day) and a low dose of ITI (50 IU/kg three times per week), the time until complete recovery was 15.5 months (IQR: 10.8–22) in the low-dose group and 10.6 months (IQR: 6.3–20.5) in the high-dose group.¹¹ Inhibitors are successfully eradicated in roughly two-thirds of patients.¹² The inhibitor relapse rate after successful ITI varies between 2.3 and 10% in most studies.^{13–15} As treatment and prophylaxis with rFVIIa or aPCC is less effective and more expensive than treatment with FVIII,⁷ morbidity among patients with inhibitors is higher and their quality of life is lower than that of patients without inhibitors.^{16,17} In patients with moderate or mild hemophilia A, the inhibitor titer may sponta-

neously decrease and become unmeasurable due to the continuing production of endogenous FVIII. However, when treatment with a (wild-type) FVIII concentrate is indicated, the inhibitor titer may rise again due to an anamnestic response, reflecting lack of sustained tolerance.¹⁸

Drawbacks of Immune Tolerance Induction

As of now, ITI is the standard treatment for patients with inhibitors. Although ITI is a safe and relatively successful inhibitor eradication strategy, the long duration and high intensity of treatment is very demanding for the usually young patients and their families, and it is very costly. In addition, effective treatment options for inhibitor patients who are refractory to ITI are lacking. New inhibitor prevention/eradication strategies are therefore urgently needed to improve patient outcomes and reduce ITI cost.

Preventing or Eradicating Antidrug Antibody Formation: What Can We Learn from Other Research Disciplines?

The problem of antidrug antibody formation is not confined to hemophilia A.¹⁹ Many biopharmaceuticals are immunogenic to a certain degree, ranging from a limited immune reaction to a major clinically relevant antibody response. For example, clinically relevant antidrug antibody formation is (or used to be) a significant problem in patients using tumor necrosis factor (TNF) inhibitors (used in rheumatic diseases),²⁰ epoetin (used for anemia in chronic renal failure),²¹ interferon- β (used in multiple sclerosis),²² alglucosidase alfa (used in Pompe's disease),²³ and pegloticase (used to treat gout).^{24,25} Most antidrug antibody research is disease-specific, and knowledge is not shared easily across research disciplines.

Promising new therapies to treat or bypass inhibitor development are also underway (e.g., engineered FVIII-specific regulatory T cells).²⁶ However, these novel therapies are still out of reach for the near future. There is a need for alternative treatment strategies that can be implemented today rather than sometime in the future (i.e., that make use of therapeutics that are currently in the market).

Over the last years, many different antidrug antibody prevention or eradication strategies (mainly using immunomodulatory agents) have been investigated in patients with disorders other than hemophilia A. This review therefore aims to compile information on the efficacy and safety of these different strategies and to contemplate whether knowledge from these other fields might inspire novel treatment strategies for hemophilia A patients.

Prevention of Antidrug Antibody Formation: What Is Already Known in Hemophilia A

In general, risk factors for inhibitor development can be divided into genetic risk factors such as FVIII genotype, ethnicity, and human leukocyte antigen type and genetic polymorphisms that encode proteins involved in the immune system, such as IL-10 and TNF- α .²⁷ In addition, there are treatment-related risk factors such as the intensity of FVIII treatment, the

frequency of FVIII exposure, FVIII dose, exposure to FVIII during surgery, prophylactic versus on-demand treatment, and the specific type of FVIII product used.²⁸ As genetic risk factors for inhibitor formation (such as FVIII genotype) are immutable, strategies to prevent inhibitor formation have focused on influencing treatment-related risk factors.

A single-arm study published in 2009 evaluated the effect of a treatment regimen that aimed to minimize the risk of inhibitor formation.²⁹ The treatment regimen consisted of early initiation of prophylaxis and minimizing exposure to “danger signals” (due to trauma, surgery, infection, vaccination, etc.) during FVIII infusion. Surprisingly, only 1/26 (3.8%) patients on this modified treatment regimen developed inhibitors compared with 14/30 (47%) patients in the control group. These results were not replicated in a follow-up study (the EPIC [Early Prophylaxis Immunologic Challenge] study) as 8/19 (42.1%) patients on the same protocol developed an inhibitor.³⁰

As mentioned earlier, there are several novel therapeutics that could be used for preventing inhibitor formation in high-risk patient groups, some of which are currently being investigated in patients with hemophilia A. These novel therapeutics include an anti-tissue factor pathway inhibitor antibody (concizumab), a bispecific antibody against FIXa/FX that mimics the function of FVIII (emicizumab), an rFVIIa product with enhanced half-life due to fusion with albumin (rFVIIa-FP), and a short interfering RNA molecule that inhibits production of antithrombin (fitusiran). As of yet, none of these therapeutics have received market approval by the U.S. Food and Drug Administration or the European Medicines Agency.³¹

Several animal studies using FVIII-deficient mice have found that a short course of treatment with rapamycin,³² anti-CD20 therapy,³³ anti-CD3 therapy,³⁴ or dexamethasone³⁵ significantly prevented inhibitor formation, even after cessation of the immunomodulatory agent. As of yet, no human studies have evaluated inhibitor prevention with these immunomodulatory agents in patients with hemophilia A.

Prevention of Antidrug Antibody Formation: What Is Known from Other Diseases

Most evidence on the prevention of antidrug antibody formation comes from patients with Pompe's disease and patients with rheumatoid arthritis (RA) or inflammatory bowel disease (IBD). The following paragraphs will review the available evidence on the efficacy and safety of antidrug antibody prevention strategies in these patient groups.

Antibodies against Recombinant Human Acid α -Glucosidase in Patients with Pompe's Disease

There is very limited experience with immunomodulatory therapy to prevent antidrug antibody formation in very young pediatric patients. Pompe's disease (also known as glycogen storage disease type II) is an autosomal recessive lysosomal storage disease caused by a deficiency of the lysosomal enzyme acid α -glucosidase (GAA), which leads to accumulation of glycogen in the lysosome. The clinical phenotype is a spectrum that ranges from the classic infantile

phenotype (the most severe form in which progressive cardiac hypertrophy is always present) to the late onset “childhood” and “adult” phenotypes.³⁶

The overall incidence of Pompe's disease is roughly 1:40,000 for all types^{37,38} and 1:138,000 for patients with the classic infantile phenotype.³⁷ Overall, antidrug antibody formation is especially problematic in patients with the classic infantile phenotype and less so in patients with the adult-onset phenotype.^{39,40}

The classic infantile form has a more severe and rapidly progressing clinical course than the late-onset childhood and adult forms and is associated with hypertrophic cardiomyopathy, muscle weakness, and respiratory distress. In general, untreated patients with classic infantile Pompe's disease die within the first year of life.^{41,42}

Enzyme replacement therapy with recombinant human acid GAA (rhGAA) is the only available treatment option. In patients with classic infantile Pompe's disease enzyme replacement therapy is initiated as soon as patients are diagnosed to prevent further clinical deterioration. Roughly, 66 to 75% of patients with classic infantile Pompe's disease have some residual GAA production (cross-reactive immunological material [CRIM]-positive patients), whereas 25 to 33% produce no GAA at all (CRIM-negative patients).^{23,43,44} Being CRIM-negative is strongly associated with a low therapeutic response to enzyme replacement therapy.²³ Several studies have shown that the majority (> 90%) of patients with infantile Pompe's disease develop antidrug antibodies, regardless of CRIM status.^{23,44,45} Antibody testing is usually performed using an enzyme-linked immunosorbent assay (ELISA). High-titer antibodies against rhGAA occur more commonly in CRIM-negative patients⁴⁴ and are associated with a poor therapeutic response to enzyme replacement therapy. Compared with patients with low-titer antibodies, patients with high-titer antibodies have worse clinical outcomes in terms of overall survival, ventilator-free survival, left ventricular mass index, and the Alberta Infant Motor Scale.⁴⁶

Antidrug Antibody Formation in Other Lysosomal Storage Disorders

Antidrug antibody formation also occurs in other lysosomal storage disorders such as Gaucher's disease and Fabry's disease. Gaucher's disease is an autosomal recessive disorder in which the enzyme glucocerebrosidase is deficient, leading to accumulation of glucocerebroside in the lysosomes of cells (mainly macrophages). The most common clinical manifestations are anemia, thrombocytopenia, hepatosplenomegaly, and various manifestations of bone disease.⁴⁷ Roughly 15% of patients with Gaucher's disease develop IgG-antibodies against glucocerebrosidase replacement therapy.⁴⁸ Over time, most patients (90%) are tolerized to glucocerebrosidase.⁴⁹ Cases of patients with sustained neutralizing antibody activity that impacts clinical efficacy are extremely rare.⁵⁰

Fabry's disease is an X-linked disorder in which the enzyme α -galactosidase A is deficient, leading to accumulation of globotriaosylceramide in cells. Clinical manifestations during childhood include neuropathic pain and angiokeratoma. In later life, renal, cardiac, and cerebral manifestations of the

disease become more prominent.⁵¹ Around 73% of men and 12% of women with Fabry's disease develop IgG-antibodies against α -galactosidase A replacement therapy. Males with Fabry's disease have less residual enzyme activity compared with females (because Fabry's disease is X-linked), which leads to higher rates of antidrug antibody formation in males. Antidrug antibody formation seems to negatively influence biochemical parameters in the blood and urine.^{52,53} The association between antidrug antibody formation and clinical outcomes is less clear.^{53,54} This is, in part, caused by the lack of a uniform assay methodology to detect antidrug antibodies and the limited effectiveness of enzyme replacement therapy in this progressive disorder.⁵³ Overall, antidrug antibody prevention/eradication strategies are very rarely applied in patients with Gaucher's disease or Fabry's disease. The following paragraph will focus on antidrug antibody prevention strategies in patients with Pompe's disease.

Overview of Antidrug Antibody Prevention Strategies in Pompe's Disease

Several small studies (mostly case reports or case series) have evaluated antidrug antibody prevention strategies; most of the CRIM-negative patients with Pompe's disease in these studies were treated with immunomodulatory agents at the start of enzyme replacement therapy. Because Pompe's disease is a progressive disorder, antidrug antibody formation (that renders enzyme replacement therapy ineffective) leads to irreversible damage. The prevention of antidrug antibodies would expectedly lead to better outcomes. The four largest case-series, all published between

2013 and 2017, included 38 CRIM-negative patients who underwent immunomodulatory therapy to prevent antidrug antibody formation (►Table 1).

In 2013, a case series⁵⁵ was published that evaluated immunomodulatory therapy in four CRIM-negative patients with Pompe's disease at the start of enzyme replacement therapy (►Table 1). Three patients received an initial cycle of rituximab and maintenance therapy with rituximab, sirolimus, and intravenous immunoglobulins (IVIGs). One patient received an initial cycle of rituximab and maintenance therapy with mycophenolate and IVIG. IVIG was given to provide passive immunity during the period of B-cell depletion due to rituximab. In total, one patient developed high-titer anti-rhGAA antibodies. This was the patient who received maintenance therapy with mycophenolate. The other three patients remained antibody-negative until the end of the follow-up. Because these patients received maintenance rituximab every 12 weeks, B-cell recovery (defined as having B cells within normal range after B-cell depletion) was not observed in these patients. No immunomodulation-related adverse events were reported, with the exception of one patient who experienced multiple viral respiratory tract infections during treatment with immunomodulatory agents. Because of prolonged B-cell depletion, patients were not vaccinated during the study (except with seasonal influenza vaccine).

That same year, another case series was published, which reported on seven CRIM-negative patients with Pompe's disease who received immunomodulatory therapy at the start of enzyme replacement therapy to prevent antidrug antibody formation (►Table 1).⁵⁶ The treatment regimen

Table 1 Overview of largest case series evaluating immunomodulatory therapy in CRIM-negative patients with classic infantile Pompe's disease

First author (year of publication)	N	Median age (range) at start of treatment	Treatment	Median follow-up (range)	Negative antibody status	Positive antibody status, median peak titer (range)	Suspected treatment-related infections	Antidrug antibody assay
Elder et al (2013) ⁵⁵	4	7 mo (2–8)	Initial rituximab cycle (375 mg/m ² per week for 3 wk or two doses of 750 mg/m ² 10–14 d apart; maintenance rituximab (375 mg/m ² every 12 wk) Sirolimus (initial dose 0.6–1 mg/m ² per day) or mycophenolate (300 mg/m ² per day); monthly IVIG (initial dose 500–1,000 mg/kg).	27.8 m (11–36)	3/4	1/4, titer not reported	1/4	Enzyme-linked immunosorbent assay
Banugaria et al (2013) ⁵⁶	7	3.5 mo (0.4–6.5)	Rituximab (IV, 375 mg/m ² weekly for 4 wk) Methotrexate (SC, 0.4 mg/kg, three doses per week for 3 wk) IVIG (400–500 mg/kg, monthly for 5–6 mo)	16.1 m (10.6–23.2)	4/7	3/7, 1:6,400 (1,600–6,400)	1/7	Enzyme-linked immunosorbent assay and confirmation using radioimmuno-precipitation, performed by product manufacturer
Broomfield et al (2016) ⁵⁷	8	4.3 mo (0–6.7)	Rituximab (IV, weekly for 4 wk, dose not reported) Methotrexate (SC, 3 d per week for 6 wk, dose not reported)	Not reported	7/8	1/8, 1:12,800	Not reported	Performed by the product manufacturer, exact methodology not reported
Kazi et al (2017) ⁵⁸	19	3.4 mo (0.1–10.9)	The ITI cycle consisted of rituximab, methotrexate, and IVIG. Exact dosing was not reported but was probably similar to the study by Banugaria et al	24.2 mo (6–100.2)	8/19	11/19, 1:6,400 (200–51,200)	4/19	Performed by the product manufacturer, exact methodology not reported

Abbreviations: CRIM, cross-reactive immunological material; IV, intravenous; IVIG, intravenous immunoglobulin; ITI, immune tolerance induction; SC, subcutaneous.

used (total duration: 5 weeks) consisted of rituximab and methotrexate; in addition, IVIG was administered. In total, four out of seven patients were antibody negative until the end of follow-up. The period between B-cell depletion with rituximab and antibody measurement might have been too short (3.5 months) to assess the effect of treatment in one patient who was antibody-negative. Furthermore, B-cell recovery was not measured at all in another antibody-negative patient. These patients were compared with a historical cohort of 11 CRIM-negative patients who were treated with enzyme replacement therapy alone. Compared with patients treated with immunomodulatory therapy, all patients treated with enzyme replacement therapy alone developed anti-rhGAA antibodies during follow-up. In addition, these patients had significantly worse clinical outcomes (ventilator-free survival and left ventricular mass index) than patients who were treated with immunomodulatory therapy. One patient developed a possible immunomodulation-related infection and had to be hospitalized.

In 2016, a retrospective analysis reported on 13 CRIM-negative patients from the United Kingdom, of whom 8 were treated with rituximab and methotrexate at the start of enzyme replacement therapy to prevent antidrug antibody formation (►Table 1).⁵⁷ One out of eight CRIM-negative patients treated with immunomodulatory therapy developed intermediate-titer anti-rhGAA antibodies (peak titer was 1:12,800 at 8 months). The remaining seven patients remained antibody-negative during follow-up. B-cell recovery after initial treatment with rituximab did not occur in one patient who remained antibody-negative during follow-up. Furthermore, it is unclear if the follow-up period after B-cell depletion with rituximab was long enough to assess the effect of treatment in the other six patients who remained antibody-negative during follow-up. Another five CRIM-negative patients did not receive immunomodulatory therapy, and of these, only two were tested for antibodies. Both patients had high-titer anti-rhGAA antibodies (peak titer was 1:204,000). Survival was higher among CRIM-negative patients receiving enzyme replacement therapy and immunomodulatory therapy when compared with CRIM-negative patients treated with enzyme replacement therapy alone. Information about adverse events during immunomodulatory therapy was not reported.

A study published in 2017 evaluated immunomodulatory therapy to prevent antidrug antibody formation at the start of enzyme replacement therapy in a larger cohort of 19 CRIM-negative patients (►Table 1).⁵⁸ The treatment regimen consisted of rituximab, methotrexate, and IVIG. Eight patients never developed antibodies. There was B-cell recovery after depletion with rituximab in all these patients. Similar to the previous study, it was unclear if the follow-up period after B-cell depletion with rituximab was long enough to assess the effect of treatment. Seven patients had low antibody titers at the end of follow-up (defined as titers $\leq$ 1:6,400). The remaining four patients developed intermediate-to-high antibody titers. These patients were compared with a historical cohort of 11 CRIM-negative patients who were treated with enzyme replacement therapy alone. All patients treated with enzyme replacement therapy alone developed anti-rhGAA antibodies

during follow-up. In addition, these patients had significantly worse survival than patients who were treated with immunomodulatory therapy. Four patients who were treated with immunomodulatory therapy developed a serious bacterial infection.

Taken together, 22 (58%) out of 38 CRIM-negative pediatric patients with classic infantile Pompe's disease who were treated with rituximab, methotrexate, and IVIG to prevent antidrug antibody formation did not develop anti-rhGAA antibodies. In comparison, literature has shown that virtually all (>90%) CRIM-negative pediatric patients with classic infantile Pompe's disease develop anti-rhGAA antibodies.^{23,44,45} In patients with Pompe's disease, it seems that concomitant immunomodulatory therapy for a short period of time (5–6 weeks) at the start of enzyme replacement therapy is effective in preventing antidrug antibody formation in a large proportion of patients. In addition, long-term follow-up results indicate that these patients maintain tolerance to rhGAA after cessation of immunomodulatory therapy. The treatment duration was very short, causing minimal interruption to the vaccination schedule, whereas the rate of adverse effects (such as opportunistic infections) was minimal.

In total, 6 out of 30 CRIM-negative patients with Pompe's disease included in the studies by Elder et al,⁵⁵ Banugaria et al,⁵⁶ and Kazi et al⁵⁸ developed a serious bacterial or viral infection during treatment with rituximab and methotrexate. Apart from infections, no other serious adverse events were reported in these studies. Information about adverse events was not reported for the eight patients included in the study by Broomfield et al.⁵⁷

Due to suppression of the immune system, the treatment regimen can reduce the response to pediatric vaccinations and may cause severe complications if a live vaccine is administered. Several studies in which rituximab was administered to patients with Pompe's disease withheld vaccination and resumed schedule after normalization of the CD19 count (which was used as a marker for B-cell recovery); this took roughly 3 to 6 months after ending the treatment regimen.^{56,58}

The reported studies had several limitations; the studies evaluating antidrug antibody prevention strategies in patients with Pompe's disease were very small and consisted of case series (due to the rarity of this disorder). Furthermore, the immunomodulatory treatment protocols varied between patients in some studies; some patients underwent several cycles of the same immunomodulatory treatment protocol, and other patients received modified versions of the protocol. The median follow-up time may have been too short to adequately assess long-term tolerance to rhGAA. For example, some patients with short-follow-up time may have been antibody-negative, months after cessation of immunomodulatory therapy, due to the lingering immunosuppressive effect of the treatment on B cells. Lastly, the antibody assay methodology was not uniform and sometimes not reported at all, complicating comparisons between studies and pooling results. Given the aforementioned drawbacks, the results of these studies should be interpreted with caution.

Prevention of Antidrug Antibody Formation in Pompe's Disease: Which Treatment Strategies Could Be Considered in Hemophilia A

A short course of treatment with rituximab, methotrexate, and IVIG was enough to prevent antidrug antibody formation and induce immune tolerance in CRIM-negative patients with classic infantile Pompe's disease. However, roughly 20% of patients developed a serious bacterial or viral infection. In hemophilia A, far less patients develop clinically relevant antidrug antibodies, and the consequences of antidrug antibody formation are not as severe. Hemophilia A patients who develop inhibitors can be treated with ITI, which has fewer side effects than treatment with immunomodulatory agents. In addition, patients who are refractory to ITI can still be treated with bypassing therapy. Therefore, in pediatric patients with hemophilia A, the benefits of this treatment protocol do not outweigh the potential risks due to adverse events (mainly infections).

Antibodies against TNF Inhibitors in Patients with Rheumatoid Arthritis or Inflammatory Bowel Disease

Anti-TNF monoclonal antibodies (TNF inhibitors) such as infliximab or adalimumab are often used in RA as a second-line agent when treatment with nonbiologic disease-modifying antirheumatic drugs (DMARDs) such as methotrexate fails. TNF inhibitors and DMARDs such as azathioprine are also used to treat patients with severe IBD (i.e., Crohn's disease or ulcerative colitis).

Large proportions of patients with RA or IBD develop antibodies against TNF inhibitors. However, due to heterogeneity in assay methodology, reported incidence rates of antidrug antibody formation vary widely. Most studies used ELISAs or radioimmunoassays (RIAs).⁵⁹ One meta-analysis that included 2,350 patients with a variety of chronic inflammatory diseases using infliximab reported that 20.8% of patients had anti-infliximab antibodies.²⁰ These antibodies also reduce the efficacy of these drugs.²⁰ Methotrexate and azathioprine are primarily used to treat disease activity in patients with RA or IBD. The immune response against TNF inhibitors may be mitigated by these drugs when they are used in combination.

Overview of Antidrug Antibody Prevention Strategies in Rheumatoid Arthritis/Inflammatory Bowel Disease

Several randomized controlled trials (RCTs) have evaluated the effect of TNF inhibitor monotherapy versus combined therapy consisting of a TNF inhibitor and methotrexate/azathioprine on antidrug antibody formation in patients with RA or IBD. Most large RCTs report a significant decrease in the incidence of antidrug antibody formation with concomitant use of methotrexate or azathioprine (►Table 2).^{60–64} Antidrug antibody formation was assessed using ELISA, RIA, or electrochemiluminescence immunoassay. All RCTs presented in ►Table 2 only included patients who had not been previously treated with the specific TNF inhibitor that was used. Similarly, a systematic review and meta-analysis that included both observational and interventional studies ($n = 2,611$) estimated that

concomitant treatment with DMARDs, mainly methotrexate and azathioprine, significantly prevented the risk of antidrug antibody formation in patients with a variety of chronic inflammatory diseases using a TNF inhibitor (odds ratio: 0.32; 95% confidence interval: 0.25–0.42).⁶⁵

Thus, strong evidence exists that methotrexate and azathioprine significantly prevent antidrug antibody formation in antibody-negative patients with RA and IBD. However, there were no methodologically sound comparative studies that assessed whether the immunomodulatory effect persisted after cessation of methotrexate or azathioprine.

Prevention of Antidrug Antibody Formation in Rheumatoid Arthritis and Inflammatory Bowel Disease: Which Treatment Strategies Could Be Considered in Hemophilia A

It is very likely that methotrexate or azathioprine would be able to prevent antidrug antibody formation in hemophilia A patients. Compared with the treatment protocol consisting of rituximab, methotrexate, and IVIG currently used to treat antidrug antibody formation in CRIM-negative patients with Pompe's disease, using only methotrexate or azathioprine would have a more favorable safety profile. However, whether immune tolerance to FVIII would persist after cessation of immunomodulatory therapy with methotrexate or azathioprine remains unknown.

The patients in the aforementioned studies received doses of methotrexate that were high enough to produce a therapeutic response. It is possible that a lower dose, with a reduced risk of adverse events, could be enough to prevent antidrug antibody formation. Based on the studies evaluating patients with Pompe's disease, these immunomodulatory agents would not have to be administered indefinitely. A short course of methotrexate at treatment initiation (e.g., during the first 10–20 exposure days to FVIII) could be sufficient to prevent antidrug antibody formation and induce immune tolerance. However, studies evaluating methotrexate in very young pediatric patients are lacking. The mean age of patients included in each study reported in ►Table 2 varied from 29 to 50.6 years. In contrast, most severe hemophilia A patients who develop an inhibitor do so at the age of 1 to 2 years old.⁶ Nevertheless, this treatment strategy could be a target for further investigation in patients at a high risk of developing persistent inhibitors. In this case, an accurate model to predict the risk of persistent inhibitor development would be needed.

Eradiation of Antidrug Antibodies: What Is Already Known in Hemophilia A

Inhibitor eradication strategies using immunomodulatory agents have been tried since the 1970s.^{66,67} Nowadays, immunomodulatory agents are mostly used as second-line therapy in patients who have already failed ITI.⁶⁸ The agents are generally administered as adjunctive therapy in combination with ITI. One of the most well-known examples is the

Table 2 An overview of the largest RCTs evaluating the effect of concomitant immunomodulation on antidrug antibody formation against TNF inhibitors in patients with rheumatoid arthritis or inflammatory bowel disease

Study author, year of publication, (disease)	Treatment	Age	Follow-up	Antibody-positive ^a /total, N (%)	Antidrug antibody assay
Colombel et al, 2010 (Crohn's disease) ⁶⁰			30 wk		Enzyme-linked immunosorbent assay
Infliximab	Intravenous infliximab, 5 mg per kg at wk 0, 2, 6, and then every 8 wk	Median: 35 y (IQR: not reported)		15/103 (14.6%)	
Infliximab + azathioprine	Intravenous infliximab, 5 mg per kg at wk 0, 2, 6, and then every 8 wk Oral azathioprine, 2.5 mg per kg daily	Median: 34 y (IQR: not reported)		1/116 (0.9%)	
Panaccione et al, 2014 (ulcerative colitis) ⁶¹			16 wk		Assay not reported
Infliximab	Intravenous infliximab, 5 mg per kg at wk 0, 2, 6, and 14	Mean: 38.5 y (SD: 12.7)		7/37 (19%)	
Infliximab + azathioprine	Intravenous infliximab, 5 mg per kg at wk 0, 2, 6, and 14 Oral azathioprine, 2.5 mg per kg daily	Mean: 38 y (SD: 12.2)		1/31 (3%)	
Matsumoto et al, 2016 (Crohn's disease) ⁶²			26 wk		Radioimmunoassay
Adalimumab	Subcutaneous adalimumab, 160 mg at week 0, 80 mg at wk 2 and then 40 mg every other week	Mean: 29 (SD: 12)		10/76 (13.2%)	
Adalimumab + azathioprine	Subcutaneous adalimumab, 160 mg at wk 0, 80 mg at wk 2, and then 40 mg every other week Oral azathioprine, maximum 100 mg daily (dose escalation from 25 or 50 mg daily to 100 mg daily during the first 4 wk)	Mean: 32 (SD: 12)		3/75 (4%)	
Emery et al, 2009 (rheumatoid arthritis) ⁶³			24 wk		Electrochemiluminescence immunoassay
Golimumab	Subcutaneous golimumab, 100 mg once monthly	Mean: 48.2 (SD: 12.85)		14/104 (13.5%)	
Golimumab + methotrexate	Subcutaneous golimumab, 50 or 100 mg once monthly Oral methotrexate, 20 mg per week (dose escalation from 10 mg per week during the first 8 wk)	Mean: 50.6 (SD: 11.58)		6/211 (2.8%)	
Kremer et al, 2010 (rheumatoid arthritis) ⁶⁴			48 wk		Enzyme-linked immunosorbent assay
Golimumab	Intravenous golimumab, 2 mg per kg or 4 mg per kg every 12 wk	Mean: 49.2 (SD not reported)		17/194 (9%)	
Golimumab + methotrexate	Intravenous golimumab, 2 mg per kg or 4 mg per kg every 12 wk Oral methotrexate, 15–25 mg per week	Mean: 49.6 (SD not reported)		10/299 (3%)	

Abbreviations: IQR, interquartile range; RCTs, randomized controlled trials; SD, standard deviation; TNF, tumor necrosis factor.

Note: All patients were antibody-negative at baseline.

^aPositive for antibodies according to the study's definition.

Malmö protocol, which consists of extracorporeal immunoadsorption followed by cyclophosphamide and IVIG in combination with high-dose ITI.⁶⁹

In 2014, a systematic review assessed the effect of immunomodulatory agents (alone or in combination with ITI) on inhibitor eradication success rates.⁶⁸ In total, 46 case reports or case series were included, comprising 208 patients. Complete recovery was defined as having a negative inhibitor titer, and having normalized pharmacokinetic parameters was not mandatory. In most cases, immunomodulatory agents were administered concomitantly with ITI. Many

patients had previously failed first-line treatment with ITI and had high-peak inhibitor titers.

Most patients were treated with cyclophosphamide (alone or in combination with other drugs) with a complete recovery rate of 40 to 44%. The second most used immunosuppressive agent was rituximab (alone or in combination with other drugs), which was associated with a complete recovery rate of 40 to 63%. As most patients failed previous ITI and had a poor prognosis for treatment success, the aforementioned success rates are quite good. However, as case reports and case series with positive results are far more likely to be published,⁷⁰ the

published recovery rates are most likely an overestimation of the true recovery rate. In addition, it was unclear if the follow-up time was long enough to accurately assess the relapse rate for most patients.

Overall, current evidence on the effectiveness of ITI in combination with an immunomodulatory agent such as rituximab or cyclophosphamide is inconclusive because of small studies with methodological limitations.^{68,71} As far as we know, no randomized studies have been performed.

Very few studies have evaluated rituximab monotherapy; this treatment could be useful due to the low costs of treatment when compared with the high cost of ITI.⁷² The largest study is a noncomparative trial from 2014 in which the effectiveness of monotherapy with rituximab was studied in 16 inhibitor patients with inhibitor titers > 5 BU (13 patients had failed previous ITI).⁷³ Only 3 out of 16 patients had a drop in inhibitor titer below 5 BU during follow-up and persistent tolerance after rechallenge with FVIII. These results suggest that inhibitor eradication with rituximab monotherapy is not as good as ITI. However, the treated group consisted of patients with a poor prognosis who failed ITI.

Eradiation of Antidrug Antibodies: What Is Known from Other Diseases

Antidrug antibody eradication strategies have been extensively described for patients with antibody-mediated pure red aplasia due to epoetin use and for multiple sclerosis patients with antibodies against interferon- β . The following paragraphs will review the available evidence on the efficacy of antidrug antibody eradication strategies in these two patient groups.

Antibody-Mediated Pure Red Cell Aplasia Due to Epoetin Use in Patients with Chronic Kidney Disease

Antibody-mediated pure red cell aplasia (PRCA) is a rare but severe side effect of treatment with epoetin (recombinant human erythropoietin) in patients with reduced production of endogenous erythropoietin, which is most often due to severe chronic kidney disease (CKD). PRCA is caused by the formation of antibodies against epoetin, which also cross-react with endogenous erythropoietin. This leads to profound anemia, a very low reticulocyte count and very low levels of erythroid precursors in the bone marrow. Antibody-mediated PRCA bears some similarity to inhibitor formation in mild hemophilia A; in both conditions, antibodies against an exogenous protein (FVIII/epoetin) cross-react with the endogenous protein (FVIII/erythropoietin). The most commonly used assays to detect antidrug antibodies are radioimmunoprecipitation assays and ELISA. Testing for neutralizing antibodies using an assay that measures in vitro inhibition of epoetin activity by antibodies is available but rarely used.⁷⁴

Overview of Antidrug Antibody Eradication Strategies in Antibody-Mediated Pure Red Cell Aplasia

Around 200 cases of antibody-mediated PRCA occurred between 1998 and 2004 and almost all were associated with the use of a particular epoetin product (tradename:

Eprex). The increased immunogenicity of this product during this time period was likely due to a formulation change.⁷⁵

A retrospective analysis⁷⁶ evaluated the long-term outcome (median follow-up: 9 months) of 170 patients with antibody-mediated PRCA due to epoetin use. Out of these 170 patients, 19 patients received a renal transplant (with concomitant immunosuppression) and 89 nontransplant patients received immunosuppressive treatment to eradicate antidrug antibodies, whereas 62 patients did not receive any treatment. In total, 44 out of 89 (49%) nontransplant patients who were treated with immunosuppressive agents to eradicate antidrug antibodies achieved hematological recovery. In comparison, only 1 out of 62 (2%) patients who received no immunomodulatory treatment achieved hematological recovery. Hematological recovery was defined as having ≤ 1 red blood cell transfusion per month, hemoglobin levels ≥ 80 g/L (8 g/dL), and a reticulocyte count $> 20 \times 10^9/L$. The specific type of antidrug antibody assay(s) used and the immunosuppressive treatment that patients underwent were not accurately reported. However, the authors report that most of the more recently diagnosed patients were treated with prednisone (alone or in combination with IVIG), cyclophosphamide (alone or in combination with prednisone), or cyclosporine. Treatment with epoetin after hematological recovery was rare; nevertheless, 19 out of 34 (56%) patients who were rechallenged with epoetin had good clinical response to epoetin. It was not reported if patients were rechallenged with Eprex or a different epoetin product. Good clinical response to epoetin was defined as having stable hemoglobin level ≥ 80 g/L (8 g/dL) and independence from red blood cells transfusions. The best predictor of good clinical response to epoetin was a negative-antibody status at reinitiation of epoetin therapy.

In a retrospective analysis of 47 patients with PRCA, the efficacy of antidrug antibody eradication strategies was evaluated.⁷⁷ Nine patients were not treated with any kind of immunomodulatory therapy, and none of these patients recovered during follow-up (median follow-up: 12 months; IQR: 8–13). Eleven patients were treated with multiple different immunosuppressive treatment protocols (the exact type of treatments was not accurately reported), and the remaining 26 patients received one type of treatment. Three treatment regimens were most commonly used; 7 out of 8 (87%) patients treated with corticosteroids and cyclophosphamide, 4 out of 6 (67%) patients treated with cyclosporine, and 10 out of 18 (56%) patients treated with corticosteroids with/without IVIG achieved hematological recovery (**Table 3**). None of the recovered patients had a relapse during the follow-up period (duration of follow-up was not reported).

Thus, treatment with immunomodulatory therapy alone seems to be effective in eradicating antidrug antibodies in CKD patients with antibody-mediated PRCA. The highest rate of hematological recovery (87%) was achieved by using a combination of corticosteroids and cyclophosphamide.⁷⁷ Around 56% of patients with successful hematological recovery had good clinical response to epoetin.⁷⁶ The exact treatment protocols were not reported and probably varied significantly on a case-by case basis. More importantly, only a small proportion of patients were reexposed to epoetin, and the

Table 3 Overview of most commonly used immunomodulatory agents used to treat antibody-mediated pure red cell aplasia^a

Treatment ^b	Time to recovery	Hematological recovery ^c
Corticosteroids, oral, starting dose: 1 mg/kg/d Cyclophosphamide, oral, dose not reported	Median duration: 3 mo (range: 1–7)	7/8 (87%)
Cyclosporine, oral, 200 mg/d	<3 wk for all patients	4/6 (67%)
Corticosteroids, oral, starting dose: 0.5–1 mg/kg/d with (<i>n</i> = 14) or without (<i>n</i> = 4) IVIg, 0.4 mg/kg daily for 5 d every 6 wk	Median duration: 3 mo (range: 1–18)	10/18 (56%)

Abbreviation: IVIg, intravenous immunoglobulin.

^aTable adapted from Verhelst et al.⁷⁷

^bSome patients were treated with multiple different immunosuppressive regimens. Consequently, the total number of treated patients is unknown.

^cHematological recovery was defined as being transfusion-independent and having a reticulocyte count of >20,000/ μ L.

level of exposure (intensity, frequency) was not reported. Therefore, the actual success rate of the used immunomodulatory agents is not known.

Eradication of Antidrug Antibodies in Antibody-Mediated Pure Red Cell Aplasia: Which Treatment Strategies Could Be Considered in Hemophilia A

Overall, the immunomodulatory agents used to treat antibody-mediated PRCA could be considered in hemophilia A patients who are refractory to ITI. However, the reported success rates of the aforementioned antidrug antibody eradication strategies (which varied from 56 to 87%) will expectedly be lower when applied to hemophilia A patients with inhibitors. This is because inhibitors in hemophilia A patients who are refractory to ITI are expected to be more difficult to eradicate. Moreover, a proportion of patients who were initially treated successfully will have an anamnestic response to FVIII after reexposure (lowering the overall success rate even further). Therefore, these antidrug antibody eradication strategies might not be as effective in hemophilia A patients with inhibitors who are refractory to ITI.

Alternatively, these treatment options could be used to treat patients in low-resource countries as a first-line treatment strategy if ITI is not available due to the costs. However, because of poor access to antibiotics and medical care in general, a severe treatment-related bacterial infection in a pediatric patient in a low-resource country would also be more difficult to treat, and therefore the benefits of this approach are not expected to outweigh the risks.

Antibodies against Interferon- β in Patients with Multiple Sclerosis

Multiple sclerosis is an autoimmune disease that is characterized by demyelination of the spinal cord and the brain. This leads to neurologic symptoms such as motor and sensory problems, paresthesia, and cognitive impairments. Patients with relapsing-remitting multiple sclerosis (RRMS) are often initially treated with interferon- β -1a or interferon- β -1b; these products are associated with relatively high rates of antidrug antibody formation. One study evaluating 1,115 Swedish and Icelandic patients reported an overall prevalence of neutralizing antidrug antibodies of 32% using a Myxovirus resistance gene A (MxA) protein assay. It is

difficult to estimate the exact prevalence of antidrug antibody formation in patients using interferon- β because the assay methodologies used to detect antidrug antibody formation are highly heterogeneous.^{78,79} However, neutralizing antibodies seem to be slightly associated with a reduced therapeutic effect of these products.⁸⁰

Glucocorticoids are used to treat exacerbations of multiple sclerosis in adults because of their anti-inflammatory and immunosuppressive effects.⁸¹ Less often, monthly therapy with glucocorticoids is used with the aim of reducing long-term disability outcomes in patients with RRMS.^{82,83}

Overview of Antidrug Antibody Eradication Strategies in Multiple Sclerosis Patients using Interferon- β

Between 2002 and 2009, three comparative trials (327 patients in total) have evaluated the use of monthly pulse therapy with methylprednisolone to prevent or eradicate antibody formation in patients with RRMS. At baseline, some or all patients were positive for antibodies against interferon- β . The results of these three studies are discussed in the following paragraphs and summarized in **Table 4**.

In 2002, an open-label RCT reported on 161 patients who were treated with either interferon- β -1b (*n* = 81) or interferon- β -1b in combination with methylprednisolone (*n* = 80).⁸⁴ The presence of neutralizing antidrug antibodies was assessed using a MxA protein assay. Antibody status of patients at baseline was not reported. After 12 months, 26.8% of patients treated with interferon- β -1b and 12.1% of patients treated with interferon- β -1b + methylprednisolone had one or more samples that were antibody-positive (relative reduction: 54.9%; *p* = 0.05). Although the reduction in antibody formation was significant, the number of disease relapses and the progression of disability during the first year of treatment were similar (**Table 4**).

In 2009, an RCT assessed treatment with either interferon- β -1a (*n* = 46) or interferon- β -1a in combination with methylprednisolone (*n* = 47).⁸⁵ The presence of neutralizing antidrug antibodies was assessed using an antiviral neutralization bioassay. Thirty-five percent of patients treated with interferon- β -1a alone and 26% of patients treated with interferon- β -1a + methylprednisolone were already antibody-positive at baseline. There was no significant difference in the cumulative incidence of antidrug antibody formation between groups; 30% of patients on interferon- β -1a and 23%

Table 4 An overview of studies evaluating the effect of concomitant immunomodulation with MP on antibody formation against INF- β in patients with multiple sclerosis

Study (year of publication)	Treatment	Study design	Age	Antibody-positive ^a /total, N at baseline (%)	Follow-up	Antibody-positive/total, N at study end (%)	Neutralizing antidrug antibody assay
Pozzilli et al (2002) ⁸⁴		RCT			15 mo		Myxovirus resistance gene A protein assay
INF- β	INF- β : subcutaneous INF- β -1b, 8 MIU every other day		Mean: 33.1 y (SD: 8.1)	7/81 (7%)		19/71 (26.8%)	
INF- β + MP	INF- β + MP: subcutaneous INF- β -1b, 8 MIU every other day Intravenous MP, 1,000 mg 1 $\times$ per month		Mean: 31.2 y (SD: 6.7)	7/80 (7%)		8/66 (12.1%)	
Sorensen et al (2009) ⁸⁵		RCT			96 wk		Antiviral neutralization bioassay
INF- β	INF- β : subcutaneous INF- β -1a, 44 μ g 3 $\times$ per week		Mean: 39.5 y (SD: 7.8)	16/46 (35%)		13/43 (30%)	
INF- β + MP	INF- β + MP: subcutaneous INF- β -1a, 44 μ g 3 $\times$ per week Oral MP, 200 mg on 5 consecutive days monthly		Mean: 37.8 y (SD: 7.4)	12/47 (26%)		9/39 (23%)	
Hesse et al (2009) ⁸⁶		Nonrandomized trial			6 mo		Cytopathic effect assay
Control group	–		Median: 41 y (range: 22–59)	35/35 (100%)		33/35 (94%)	
MP	MP: Oral, 500 mg on 3 consecutive days monthly		Median: 43 y (range: 27–62)	38/38 (100%)		36/38 (95%)	

Abbreviations: INF- β , interferon- β ; MP, methylprednisolone; RCT, randomized controlled trial; SD, standard deviation.

^aPositive for antibodies according to the study definition.

Note: 7, the number of patients with antibodies against interferon- β at the start of the study was not reported.

of patients on interferon- β -1a + methylprednisolone were antibody-positive after 96 weeks ($\rightarrow$ Table 4).

Lastly, a nonrandomized clinical trial evaluated if methylprednisolone could be used to restore interferon- β bioactivity in antibody-positive patients with absent in vivo response to interferon- β who had discontinued interferon- β therapy.⁸⁶ The presence of neutralizing antidrug antibodies was assessed using a cytopathic effect assay. Thirty-eight patients were treated with methylprednisolone, and 35 patients were not treated. The in vivo response to interferon- β and antibody status were similar after 6 months ($\rightarrow$ Table 4).

Overall, evidence from studies in patients with multiple sclerosis suggests that methylprednisolone is not effective in preventing or eradicating antibodies against interferon- β .

Eradiation of Antidrug Antibodies in Multiple Sclerosis: Which Treatment Strategies Could Be Considered in Hemophilia A

It seems that monthly treatment with methylprednisolone has no added benefit in terms of preventing or eradicating antibodies against interferon- β . In contrast, using oral corticosteroids to treat patients with antibody-mediated PRCA was a moderately successful strategy ($\rightarrow$ Table 3). The difference in efficacy may be partly explained by the fact that in patients with antibody-mediated PRCA, oral corticosteroids were mostly given in combination with IVIG or cyclophosphamide.

Given the aforementioned results, inhibitor eradication in hemophilia A patients using methylprednisolone alone should not be considered as it is not expected to be effective.

Conclusion

Insights gained from clinical research into antidrug antibody formation in other diseases could be helpful in devising alternative treatment strategies for inhibitor development in hemophilia A. Immunomodulatory treatment can be associated with potentially severe side effects. The benefits of this treatment, however, may outweigh the potential risks in subgroups of inhibitor patients with poor prognosis.

Conflict of Interest

None.

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